

Harsh R. Bohra

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Full-stack engineer who takes half-written specs to deployed, paying-customer products. Most recently that meant a production e-commerce platform of ~6,400 products for a UK retailer (cwtchcomfort.com) and founding engineer roles at two early-stage startups. I own the API, the data model, the integrations, the security pass, and the 2am bug. TypeScript and Python by default; Next.js, NestJS, React Native, Postgres.

EXPERIENCE

Software Engineer · Freelance

Mar 2025 – Present · Cardiff

End-to-end delivery for UK small businesses: discovery, build, deploy, handover (runs alongside startup roles).

Technicon Furnishers ("Cwtch Comfort"), a two-branch furniture retailer. Sole engineer for the full product suite at cwtchcomfort.com (Next.js 14, Vercel, production):

- **E-commerce platform, ~6,400 products across 12 supplier brands:** faceted shop with mega-menu categories, photography-led product pages with a finish switcher, a two-axis sofa configurator, a parametric bed builder, and a wardrobe configurator with live VAT-inclusive pricing. Stripe-hosted checkout with **server-side price and minimum-quantity validation** and signature-verified webhooks; postcode-gated delivery via a KV allowlist; consent-gated GA4 with purchase tracking; Sentry across client, server and edge.
- **Reliability and performance:** orders are persisted to Upstash KV before any email is attempted, with an /admin/orders viewer and Sentry alerts, so no order is silently lost. Moved the 17.5 MB catalogue out of the client bundle (the cart hydrates through an API resolver), cutting cart first-load JS to 221 kB.
- **Supplier data integration (with supplier consent):** ETL pipelines that normalise 12 different sources (authenticated trade portals, brochure PDFs, spreadsheets) into one product schema. Highlights include a 1,015-product wholesaler catalogue with 9,968 HD images acquired with **zero errors** through idempotent, resumable passes tuned to a fragile server, and an Italian manufacturer's 192-model, 3,075-variant range parsed from a custom CMS complete with dimensional tables and schematics.
- **Operations and billing system (FastAPI + SQLite)** covering the branches' full retail lifecycle: billing, supplier order batching, goods-received, delivery scheduling, payments and cash ledger, A4 print invoices, complaints, PIN-based admin and public order tracking. 17 pytest tests plus a scripted 21-assertion HTTP end-to-end walkthrough.
- **ML in production:** an Ollama vision pipeline that auto-classifies ~8,000 product images (parallel workers, checkpointed and resumable).

Other clients: pawtrove.shop (live Shopify build, SEO score 100); static storefronts with Node/Express/Stripe checkout and email receipts.

Founding Engineer · Nextus, identity infrastructure startup

May 2026 – Present

Sole engineer on the gateway service for Nextus' identity-before-signup primitive: phone numbers act as pre-existing endpoints, with deterministic E.164 addressing, lazy identity creation on first contact, and an OTP claim state machine. Node/Express, Postgres (Prisma), Redis.

- Immutable event log (7 typed event types, correlation IDs end to end) so every state change is replayable and auditable; Stripe-style idempotency keys, Zod validation, Redis rate limits.
- A deliberate consistency model: transactional writes (identity, interaction, graph edge and event in one Prisma transaction) and eventually consistent reads (5-minute Redis cache with explicit invalidation). 131 tests including HTTP integration against real Postgres and Redis, plus a written threat model covering OTP brute-force probability and hashing hardness.

Founding Engineer · Cache Wallet, crypto wallet startup

Jan 2026 – May 2026

Built the first functional prototype from an abandoned codebase; delivered swap, fiat ramp, and identity features across backend (NestJS/Postgres), mobile (React Native/Expo), and smart contracts (Solidity) on 7 EVM networks; tested end-to-end with real funds pre-launch.

- Gasless swap pipeline (same-chain and cross-chain) with quote aggregation, EIP-712 signed orders, and a provider-agnostic execution layer; verified a live mainnet swap with real funds.
- Diagnosed a chain-wide swap failure (1inch Fusion ProxyFactory-derived maker addresses) by tracing SDK signature recovery, and shipped a WETH pre-wrap fix the same day.
- Pre-launch security pass: patched IDOR bugs on user-scoped endpoints, migrated to short-lived JWTs with refresh rotation and revocation, and cut dependency vulnerabilities from 60 to 8.

- Fiat on/off-ramp through Meld (Banxa, Transak, MoonPay) behind one quote/session abstraction, with an interface-driven mock layer so mobile work never blocked on provider onboarding.

Software Developer Intern · Cardiff University, School of Architecture Jun – Oct 2024
Co-developed InvestigationMap (Flask, PostgreSQL, NLP pipelines); raised sentiment-analysis throughput 25% to 2,900+ entries/sec; Agile ceremonies and code review with an interdisciplinary team.

Business Development Manager · Sri Viratra Jewellers & Finance, India Jun 2021 – Jul 2023
Assisted building the internal CRM/inventory system (ElectronJS, React, SQLite) used for 1,000+ clients, the first commercial software I shipped to non-technical users; roughly 30% fewer manual errors.

EDUCATION

MSc Computing · Cardiff University · 2:1 (2023–2024). Dissertation: "Little Investigators", a geo-emotion mapping platform delivered for the Built Environment Trust (Flask, NLP, GDPR privacy-by-design, iterative client engagement).
BA Psychology, Sociology & English Literature · Christ University (2016–2019)

TECHNICAL

Primary: TypeScript, Node.js, NestJS, Next.js, React, React Native (Expo), PostgreSQL, Redis, Python (FastAPI/Flask), Stripe.
Also: Prisma, Solidity, Ethers.js, Thirdweb, Docker, Vercel, Sentry, GitHub Actions, pytest/Jest, web scraping and PDF extraction, LLM/RAG app development (Ollama, Transformers).
Full right to work in the UK. Languages: English, Hindi, Tamil, Marwari.